

Economics G6410 - Mathematical Methods for Economists - Fall 2020
Math Camp Exam

Department of Economics
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Instructions:

This is a timed take home exam. You have 3 hours to solve the exam and return a scanned copy or a typed copy of your answers in a single PDF document to the TA (and cc the instructor). There are 5 questions in total and the full grade is 100 points. You are not allowed to discuss your answers with any other students. In case of any clarifications, explicitly state the assumptions you are making to solve the question. Kindly show your complete work. Partial credit would be awarded in case of incomplete answers. No credit would be awarded if working is not shown. Columbia's academic honor code applies.

1. **(40 points)** Are the following statements true or false? If true, provide a proof. If false, provide a counter example
 - (a) Let (X, d) be a metric space. Let $S \subset X$ be a finite set. S is closed, bounded, and compact.
 - (b) Any collection of vectors that include the zero vector cannot be linearly independent.
 - (c) The sum of two quasi-concave functions is a quasi-concave function.
 - (d) In $(\mathbb{R}^n, d_2)$, intersection of compact sets is compact and finite union of compact sets is compact.
 - (e) Let (X, d_X) and (Y, d_Y) be metric spaces, and function $f : X \rightarrow Y$. The function f is a continuous function iff $f(E)$ is open in (Y, d_Y) for any set E open in (X, d_X) .
 - (f) Every diagonalizable matrix is invertible.
 - (g) Let $\{x_n\}_{n=1}^\infty$ be a convergent sequence with limit x in $(\mathbb{R}, d_2)$, and let b be a number such that $x_n \leq b$ for all n . Then $x \leq b$
 - (h) Let (X, d) be a metric space, and let S be a closed set in (X, d) . Then a Cauchy sequence (x_n) in S must be convergent.

2. **(10 points)** Consider the following matrix

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

is it diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If yes, diagonalize it, i.e. find a 2x2 matrix P such that $A = P\Lambda P^{-1}$, where Λ is a diagonal matrix. Also, find an expression for A^n for an $n \in \mathbb{N}$. As $n \rightarrow \infty$, does A^n converge to some real matrix B ? Is the matrix A positive definite? negative definite? neither?

3. **(15 points)**. Consider the following function defined for $X_1 > 0, X_2 > 0, \dots, X_n > 0$ as

$$Y = F(X_1, X_2, \dots, X_n) = \sum_{i=1}^n \alpha_i X_i^\gamma$$

where $\alpha_i > 0, i = 1, \dots, n$ and $\gamma \neq 0$ are parameters.

- (a) Is F homogeneous? If so, state its order of homogeneity. Is F homothetic?
- (b) For what values of γ is F quasiconcave? For what values of γ is it quasiconvex?
- (c) Log-linearize the equation $Y = F(X_1, X_2, \dots, X_n)$ around a point $(Y^*, X_1^*, \dots, X_n^*)$ satisfying this equation and represent the linearised equation in a percentage deviation format, i.e., the variables should take the form $\hat{z} = \frac{z - z^*}{z^*}$.

4. **(15 points)** Consider the correspondence $\Gamma : \mathbb{R} \rightrightarrows \mathbb{R}$

$$\Gamma(x) = \begin{cases} 0 & \text{if } x \in (-\infty, -2] \\ [-1, 1] & \text{if } x \in (-2, -1] \\ 0 & \text{if } x \in (-1, 0) \\ [0, 1] & \text{if } x = 0 \\ [-1 + x, 1 - x] & \text{if } x \in (0, 1) \\ 1 & \text{if } x \in [1, \infty) \end{cases}$$

- (a) Plot the correspondence
- (b) Study the upper and lower hemicontinuity of Γ , i.e. state at what points is the correspondence upper hemicontinuous and what points is it lower hemicontinuous (No proof required)

5. **(20 points)** Consider the following problem

$$\max_{(x_1, x_2) \in \mathbb{R}^2} \alpha \ln(x_1) + \beta \ln(x_2)$$

s.t

$$\begin{aligned} p_1 x_1 + p_2 x_2 &\leq m, \\ x_1 &\geq 0, \\ x_2 &\geq 0 \end{aligned}$$

where $\alpha, \beta, p_1, p_2, m \in \mathbb{R}_{++}$ are parameters.

- (a) Does a solution exist? If yes, why?
- (b) Argue that the budget constraint binds at any possible maximum.
- (c) Show that the constraints $x_1 \geq 0$ and $x_2 \geq 0$ are non binding at the maximum.
- (d) Are there any potential maximisers where the constraint qualification would fail?
- (e) Set up the Lagrange and solve the problem.
- (f) What theorem guarantees the continuity of the functions $x_1^*(p_1, p_2, m, \alpha, \beta)$ and $x_2^*(p_1, p_2, m, \alpha, \beta)$ and why?